Unit 1: Overview

Grade Level: Middle School

Intro to Computers

Overview

This unit on computers is meant to dive into what a computer is, how it was invented, who are the key inventors, what is inside a computer.

Objectives

1. Gain knowledge about the history of computers and inventors.
2. Learn what is inside a computer including the operating system and applications.
3. Be introduced to computer programming languages and their purpose.
4. Learn about the computers the students interact with everyday.

Vocabulary

Computer – An electronic device that processes information by following instructions from hardware and software.
Hardware – The physical components of a computer that you can touch, like the CPU, keyboard, or monitor.
Software – A set of instructions that tells a computer what to do; includes applications and operating systems.
Input Device – A hardware component that sends information into a computer.
Output Device – A hardware component that displays or outputs information from a computer.
CPU (Central Processing Unit) – The "brain" of the computer that carries out instructions and performs calculations.
RAM (Random Access Memory) – The computer's short-term memory that temporarily stores data while the computer is running.
Storage – A device that holds data permanently or semi-permanently.
Motherboard – The main circuit board that connects all computer components and allows them to communicate.
Power Supply – The hardware that provides power to the computer.
Operating System (OS) – Software that manages computer hardware and software resources.
Application (App) – A type of software designed to perform specific tasks.
Peripheral – External devices connected to a computer, like printers, scanners, or game controllers.
Cloud Storage – A way of storing files online instead of on your computer's hard drive.
Network – A group of connected computers that share resources and information.
Algorithm – A step-by-step set of instructions for solving a problem or completing a task.
Binary Code – The language of computers, using only 1s and 0s to represent data.
Data – Information processed or stored by a computer.
Server – A computer that provides data and services to other computers over a network.
User Interface (UI) – The way a user interacts with a computer or software, often through buttons, menus, and icons.

Time Frame

Lesson One - Cultural/Historical Overview
Lesson Two - Input/Output Process
Lesson Three - Logic & Algorithm
Lesson Four - Understanding Data & Binary Code
Lesson Five - Why Computers Matter

Teacher Resources

Code.org / Educational Videos

https://code.org/educate/resources/videos

Short videos on different computer related information.

ComputerHistory.org / Timeline

https://www.computerhistory.org/timeline/computers/

Visual Timeline starting with Bell Labs in 1937.

Materials

- Access to the Internet
- Documents/Slideshows provided by this curriculum
- Access to Google Docs/ Slides and other related software.
- Access to websites outlined in each lesson.

Unit 1: Lesson 1 **Grade Level:** Middle School

History of Computers

Description

Students will define what a computer is, identify key hardware and software components, and understand the evolution of computers by exploring significant milestones in computing history.

Objectives

1. Definition of a computer
2. Hardware vs. Software
3. Early computing devices and pioneers
4. How computers evolved from mechanical machines to modern systems
5. The relationship between historical innovations and today's technology

Anticipatory Set (5-10 mins)

Ask students:

"What's the most important piece of technology you use every day? Could you live without it?"

Facilitate a short discussion and write their responses on the board.

Transition:

"Today we're going to explore how those devices evolved — and what makes them work."

Timeline Structure

Breakdown the lesson into daily activities. This lesson should take 3/4 classes to complete at 50 minutes per class.

Day 1 – Introduction to Historical Foundations

Present the slideshow on the evolution of computers. Have the students follow along and take a quiz at the end of the presentation. There will be an open discussion on the topic of computers at the end of class.

Day 2 – Preparation for Presentations

Introduce the assignment that is due the next class. Teach students how to research their topics. Give the students options on how they can complete their presentations.

Day 3 & 4 – Student Presentations

Each student or group will present their project on technology. Depending on how long each presentation takes will be if this lesson spills into an extra day. Give students reflections to write about their favorite fact they learned from the presentations.

Example Questions

- *Why did people invent these devices?"*
- *"How did they solve real-world problems?"*
- Use visuals and short video clips to keep students engaged.

Materials

- A computer with access to the internet
- Google docs and software that can create presentations
- Historical Slideshow
- Invention/Inventor Cards
- Project Planning sheets for brainstorming presentation forms.
- Rubric/Student Reflections
- History Quiz

Day 1: Slideshow

Notes + Additions

Remember that these are middle schoolers between the ages of 11 and 14. Keep it informative but fun and engaging.

Included in this unit is the slideshow and a separate document with the information about various people and inventions with a "fun fact." Use this to build a more tailored learning experience.

The digital form of the Timeline Quiz is on **Google Forms:**
https://forms.gle/hi6c73bqHHFUcW9a9

You may also make your own.

The Evolution of Computers: 30 minutes

Slideshow Presentation

Key topics to cover:

- **What is a Computer?** Emphasize that a computer is any device that processes information.
- **Historical Milestones:** Highlight key innovators like Charles Babbage, Ada Lovelace, Alan Turing, and Grace Hopper.
- **From Mechanical to Digital:** Discuss devices like the Analytical Engine, ENIAC, and the rise of personal computers.
- **Modern Computing:** Show examples of computers in everyday life (smartphones, cars, smart home devices, etc.).

Guided Practice: 10 mins

Activity: "Quick Timeline Quiz" (Collaborative Discussion)

- On their computers, students access an interactive quiz or digital worksheet.
- The quiz will present key computing milestones in the wrong order.
- Students collaborate in pairs to arrange them in the correct order using their notes and what they learned in the presentation.
- Regroup for a short discussion: "Which innovator or invention surprised you the most?"

Closure: 5-10 mins

Reflection Prompt: "How did learning about computing history change the way you see modern devices?"

Homework: Ask students to reflect on their favorite invention or inventor that we discussed today and write a paragraph about it.

Day 2: Presentation Prep

Notes + Additions

Included with this lesson are print outs used for this mystery box. There are also print outs for the projects. I encourage you to diversify for representation and possibly use this time to have the students choose at random the projects they will be engaging in.

Anticipatory Set (5 - 10 mins)

Activity: "Invention Mystery Box"

Present a "mystery box" (or digital version) filled with clues about different computing inventions.
Each clue represents a key invention or computing pioneer.
Examples of clues:

- A punch card → ENIAC
- A floppy disk → Early data storage
- A robot toy → Robotics programming

Students work in small groups to pull a clue, discuss what they think it represents, and brainstorm how they could build a project around it.

Transition to the next activity: Ask students, "What kind of story would you tell about this invention?"

Project-Based Learning (30 - 40 mins)

Step 1: Explore Project Types

Students will explore project options using a digital handout or printed handout. The different types of project ideas are listed as follows:

- **The Time Traveler's Report** – A TEDx Talk-style speech where students "travel back in time" to explain how the invention will change the world.
- **The Tech Detective** – A true-crime style video investigating the social impact of an invention.
- **The Digital Museum Exhibit** – A visual presentation showcasing the invention's features, history, and influence.
- **The Visual Timeline Project** – A visual timeline connecting the invention to modern technology.

The project presentations should not exceed 3:30, but not be less than 1:30. The presentation should have a good amount of information that is engaging to their peers. The information must be accurate and sourced!

Step 2: Brainstorm with Group Members

The students will be broken up into pairs or small groups to complete the brainstorming portion of the process. They'll decide on the type of project they wish to do and will be given a card with the project they will be researching.

Step 3: Project Planning/Idea Mapping

Students will be given a project planning sheet to make this process a bit smoother. On the sheet they will be able to map out:

1. The invention or inventor they are focusing on.
2. What was the problem that was solved.
3. The story that they will tell using their chosen format.
4. Start researching their project.

Step 4: Independent Work

Students will spend the remainder of the class period working on researching and refining their ideas. Take the time to walk around the classroom to ensure that everyone is able to work on their project. Be proactive.

Closure: 5-10 mins

"What project format are you most excited about, and why?"
"What's one challenge you'll need to overcome to make your project successful?"

Adaptations for Diverse Learners

Visual Supports: Use diagrams, timelines, and labeled images during the slideshow and project planning stages.

Guided Notes: Provide a partially filled outline for students who need structure.

Flexible Project Formats: Offer multiple presentation options like digital slides, posters, or recorded videos to accommodate various learning preferences.

Peer Support: Partner ELL students or students with processing needs with peers during collaborative activities.

Time Management Support: Provide clear milestones during project work to keep students on track.

Flexible Presentation Options: Allow students with anxiety to record their presentation or present privately if needed.

Alternative Reflection Methods: For students who struggle with writing, offer options like voice recordings, drawing connections on concept maps, or brief one-on-one discussions.

Day 3 & 4: Presentations

Criteria for Assessment

Were Students:

- Able to follow instructions and complete their project?
- Able to present their invention's purpose, impact, and relevance to modern technology?
- Able to create a clear, organized presentation using visual aids or storytelling elements?
- Able to communicate key ideas effectively to their peers?
- Able to reflect on what they learned and explain their project's significance?

Methods for Assessment

- Instructor Observation: During project planning, design, and presentations to assess engagement and understanding.
- Checkpoint Discussions: Asking guiding questions throughout the project process.
- Presentation Grading: Using the Pillars of Computer Science Rubric.
- Reflections: Able to engage with peers and listen to presentations.

Day 3: Opening Instructions (5 - 10 mins)

Review the project rubric with the students.

Remember to emphasize key points:

1. Why was the invention/inventor important?
2. Share one interesting fact about the invention/inventor.
3. Use visuals to engage the audience.

Project Work Time: 25-30 mins

Students should be working on their projects. While the students are working, as an educator you should be circling the room and offering guidance to the students that need help.

This is also a good time to ask questions to the students to check for understanding:

"What inspired this invention?"

"Is there a modern device that wouldn't exist if this wasn't invented?"

Presentations: 10-15 mins

- Students that are ready to present the projects are encouraged to do so.
- Encourage students to ask questions to their peers for engagement.

Day 4: Finish Presentations

- Students will use this entire class to finish presenting to the class.
- Any remaining time at the end of class will be used on reflections and a class discussion.

Possible prompts for the class discussion:

- "What's one invention that surprised you and why?"
- "What's one way technology might change in the future?"
- "How does this invention connect to the technology you use every day?"

Key Message for Students

"The inventions you're learning about didn't just change technology — they changed the way people live and solve problems. Your project is your chance to tell that story in your own unique way."

Student Assessment

Don't forget to explain the rubric and what it means to the students. This provides them clear guidelines of the assignment expectations.

California/National Computer Science Standards

This Lesson meets the following standards set by the state and board of education.
6-8.CS.1, 6-8.CS.2, 6-8.IC.20, 6-8.IC.21, 6-8.AP.17, 6-8.AP.18, 6-8.AP.15, 6-8.AP.16

Rubric	4 (Excellent)	3 (Good)	2 (Fair)	1 (Poor)
Computer Literacy / Communication	Clear, organized presentation with accurate information. The invention's purpose, function, and impact are explained effectively.	Presentation is mostly clear with minor gaps in explanation or organization.	Presentation is somewhat organized but lacks clear explanations or details.	Presentation is unclear, disorganized, or missing key information.
Impacts of Computing	Effectively explains how the invention influenced society, technology, or ethics. Strong connections to modern computing.	Explains some societal or technological impacts with minor gaps in detail.	Provides minimal explanation of the invention's impact or relevance.	No clear explanation of the invention's impact or connections to modern computing.
Design	Visuals are clear, organized, and enhance the presentation. Strong creative elements support understanding.	Visuals are present and somewhat effective, but may lack creativity or clarity.	Visuals are present but poorly organized or ineffective.	No visuals included, or visuals are confusing and unhelpful.
Collaboration	All team members contributed meaningfully. Strong evidence of teamwork and shared responsibility.	Most team members contributed equally with minor imbalances.	One or two students did most of the work; collaboration was limited.	Little evidence of collaboration; work appears disjointed.

